Quantum Computing and the Innovation of Google's Willow

New innovations, like Google's Willow processor, have made quantum computing a more probable possibility for practical solutions to many real life applications. However, to understand the changes and developments in Willow that make it significant, one needs to understand the basics of quantum computing first.

Quantum computing uses the science of quantum mechanics, the study of the behavior of matter at impossibly tiny scales. Quantum mechanics studies matter and energy at the most fundamental level in order to better understand the building blocks of nature. This field has allowed humans to understand and predict the behavior of tiny particles and how they'd behave at their scale, with extremely contrasting conditions compared to the human world. Quantum discoveries have helped mankind understand foundations of materials, biology, chemistry, and astronomy. It's also the reason for many of the technological developments seen today.

Large-scale quantum computers use quantum mechanics to tackle complex problems at rates much faster than modern computers. Quantum computers consist of quantum bits, also known as qubits. Similar to bits in traditional digital computers, qubits are the basic unit of information used to store data in a quantum computer. A classic bit can only exist in 0 or 1, but qubits can also occupy a third state known as a super position. Superpositions represent 0, 1, and all the positions in between all at once. Qubits hold a probabilistic mixture of both 0 and 1 simultaneously. This means that one qubit doesn't only represent 1 or 0, it represents a spectrum of possibilities. The flexibility of qubits then allows quantum computers to explore numerous possible solutions simultaneously. Qubits can also be entangled. This means that the state of one qubit affects the state of another simultaneously, regardless of physical distance. Entanglement

allows for quantum computers to store relationships between qubits that wouldn't be possible in classical bits. Since qubits represent a spectrum, when measured the qubit "collapses" to either 0 or 1 with probabilities determined by its state. There are various kinds of qubits that are used for different applications of quantum computing.

By manipulating information in qubits, scientists produce optimal solutions to difficult problems. Simply put, quantum algorithms and calculations are performed by forcing qubits into certain states, manipulating them, and measuring the changes in terms of a solution. A quantum algorithm specifically manipulates qubits so that multiple possibilities are tested parallely, and the most efficient solution is amplified.

Like every upcoming technology, quantum computing also has its challenges. A major challenge in quantum computing is that the quantum information is very delicate and vulnerable to errors. Qubits are very sensitive to their environment. Decoherence is the process where a quantum system loses its quantum properties and classical behavior emerges because of "harmful" environmental conditions. Quantum systems can exist only in coherent states. This includes conditions such as extremely low temperatures (fraction of a degree above absolute zero) and away from heat and electromagnetic disturbances. High energy radiation can cause unintentional changes in bits (bit flips), so mitigating these high energy impacts is very critical.

Amidst the field of quantum computing, one relevant development is the creation of Google's Willow quantum processor. One major development in Willow is that it accounts for the errors created during the process of decoherence, like mentioned above. Most quantum devices usually experience at least one failure in every thousand operations, which is way too high of a rate for the billions and trillions of operations done in applications. Quantum error correction (QEC) is the important strategy needed to get rid of these errors and consequent

failures. The idea is to distribute the quantum information of a single qubit across many entangled qubits. This can help the system tolerate some individual qubits failing without losing the entirety of the information. Willow is the first quantum processor to demonstrate that its error-corrected qubits improve exponentially as they increase in size. One common QEC method that Willow encompasses is surface code. Surface code is designed to protect delicate quantum information from errors.

The surface code is structured as a two dimensional lattice of neighboring coupled qubits. The larger the lattice, the greater the code distance, and the more errors the resulting qubit can tolerate. A surface code lattice is composed of two distinct types of physical qubits, data and measure qubits. Data qubits store the quantum state of the logical qubit while measure qubits are used to repeatedly check for errors on their neighbors without distributing the stored quantum information. Measure qubits solely exist for detection, not data storage.

Surface code error correction works using repeated cycles of detection and decoding. Quantum errors are far more complex than classic bit flip errors. The surface code is designed to protect and account for these local quantum errors as well. The measure qubits perform repeated parity checks on the data qubits they are connected to. Different measure qubits are assigned to detect different kinds of errors. When an error occurs on a data qubit, it “activates” neighboring measure qubits. The measure qubits as a result react or “light up” indicating the location and type of error that has occurred.

Then, information from these activated measure qubits is sent to a classical co processor that’s running a quantum error decoder. A classical co-processor is a critical component of fault tolerance in quantum computers. This processor handles errors in real time, by using classical software to interpret measurement data and manage the error correction process. The error

information has to be processed at the same rate that the quantum computer produces it. This could be a challenge for super conducting qubits which can have every short error correcting cycle times (as short as 1.1 microseconds). If the decoder can't keep up, a backlog of information builds up. The growing backlog can cause an exponential increase in total computational time. The error data from the measure qubits is sent to the co-processor through a low latency connection. A low latency connection is a computer network that is optimized to specifically minimize data between sending and receiving data. Inside the processor, using realtime decoding software, like the sparse blossom algorithm, common error patterns are resolved quickly. The decoder then processes the data and returns its correction. In Willow, AI plays a crucial role in the software side during error handling. It specifically assists in decoding the error information generated by the quantum chip.

One example of this in Willow, is the neural network decoder. This AI based decoder interprets the measurement data and determines the errors that are most probable to occur. Since the AI decoder is highly accurate, it helps provide much faster and accurate results.

A majority of Willow's advancements occur on the basis of new hardware and operational advancements over past processors like Sycamore. The improved hardware allows Willow to accommodate for the five-fold increase in quantum coherence times and much higher computational power. Coherence refers to the average qubit lifetime. Improvements in coherence allows for more accuracy for all the operations performed on the chip. Despite the breakthroughs that Willow has brought, the solution for a large-scale, an entirely fault tolerant quantum computer still has a long way to go.

Willow still has a decoder delay time from 50 to 100 microseconds. The decoder delay time is also expected to increase with larger lattice sizes. Experiments with repetition on Willow

have revealed that there is an error floor. An error floor is essentially a limit below which the error rate will not drop. Error floors, if manipulated accordingly, can be used to an advantage by minimizing the encoded error rate as much as possible before the error floor is reached. In Willow, initially there is dramatic exponential error suppression. Yet as the error floor was approached the error suppression factor deviated from the exponential error suppression. It seems that the error floor appears with random rare correlated error bursts. More specific causes and identification methods for the error floor would be a point of improvement in future processors.

Quantum computing holds the promise for pioneering breakthroughs in countless STEM fields. These solutions could take form in terms of discovering new medicines, designing more efficient power sources, and the development of materials with certain enhanced properties. Though Willow provides a significant feat in quantum computing, there are still many more developments that could be made in the near future, holding the potential for much more to come.

Citations

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